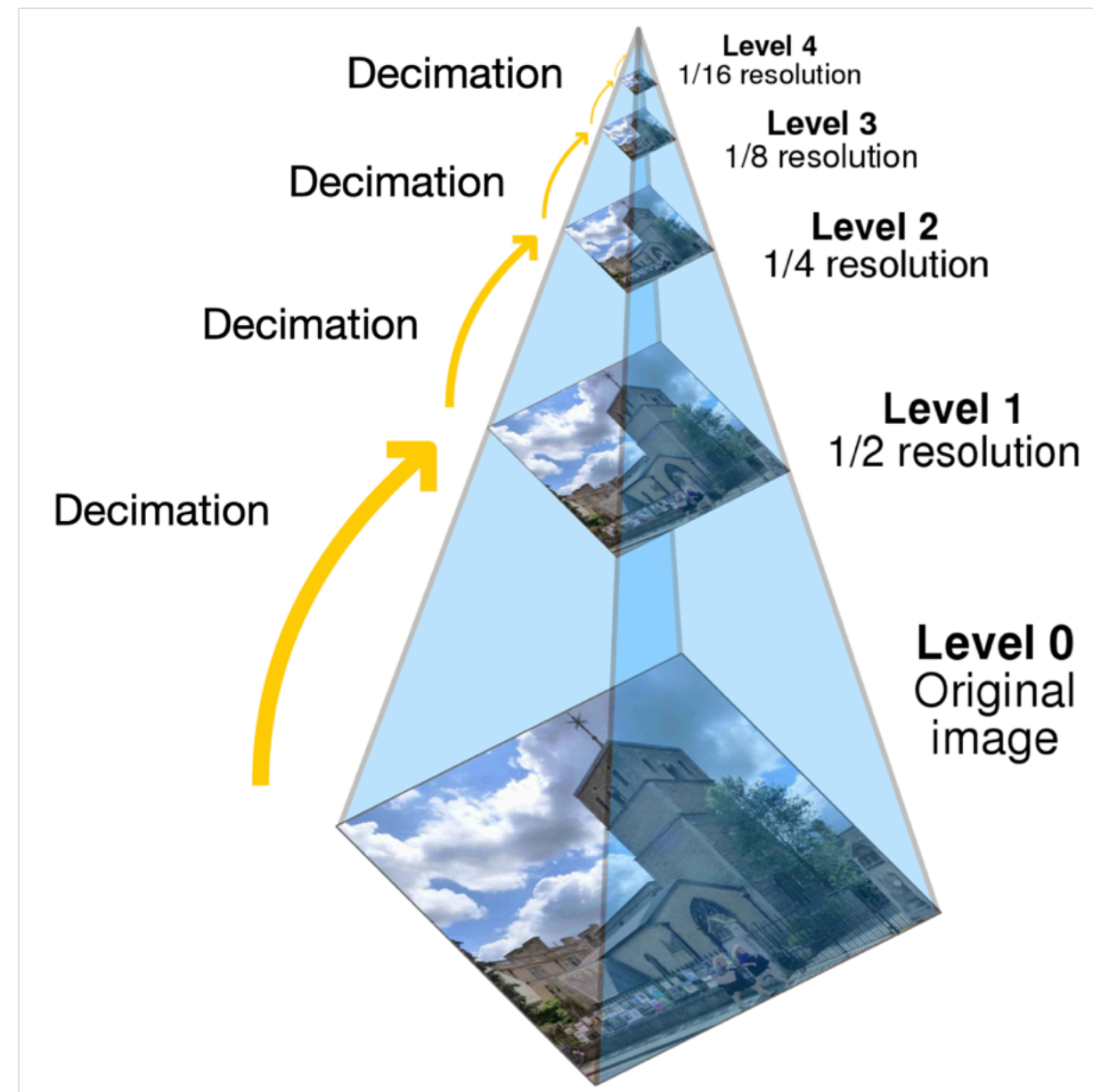


Part I : multilevel paradigm



■ Sign
■ Exponent
■ Significand

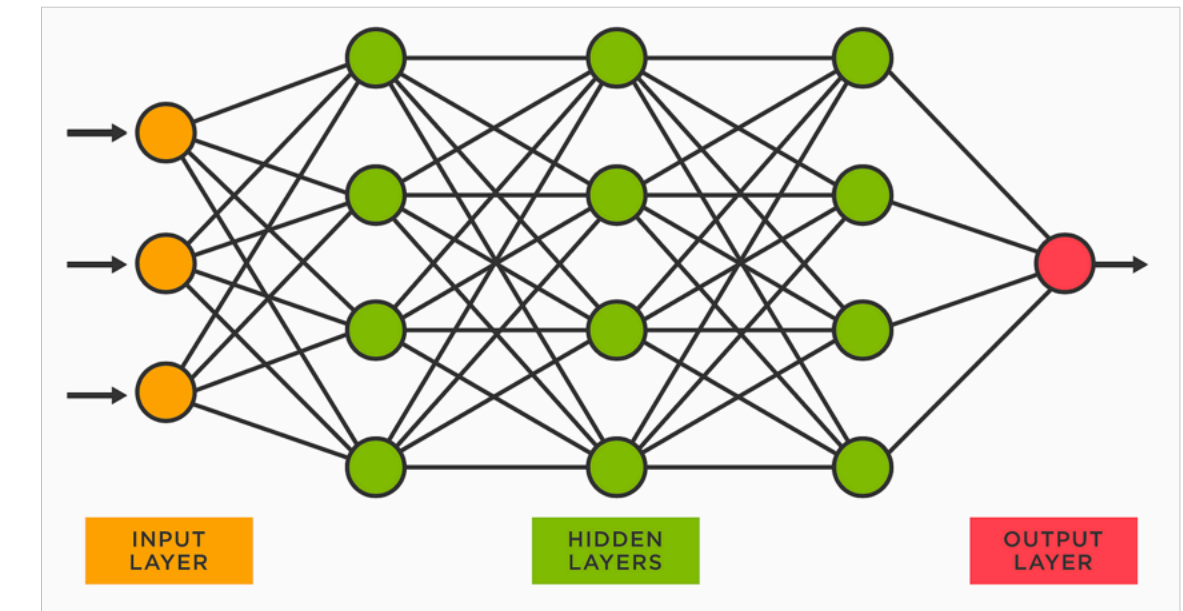
**Part II :
Image
restoration**

**Part III :
Learning
problems**

**Part V :
Arithmetic
precision**

**Part IV :
Matrix
factorization**

$$\min_{\theta} f(\theta) = \sum_{i=1}^m f_i(\theta)$$



$$\text{NN}_{\theta} = \sigma(W_L \sigma(W_{L-1} \dots \sigma(W_2 \sigma(W_1 x))))$$

